

P-Adic Feynman and String Amplitudes

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Abstract. We derive an explicit representation for *p*-adic Feynman and Koba–Nielsen amplitudes and we briefly outline the connection between the scalar models of *p*-adic quantum field theory and Dyson’s hierarchical models.

1. Introduction

As we have shown previously (see [1], submitted to “Theoretical and Mathematical Physics” in May 1987), the scalar models of the field theory over the *p*-adic field Q_p are the natural continuous analogs of Dyson’s hierarchical models (see [2–5]). More precisely, the discretization of the field theory over Q_p on the hierarchical lattice of *p*-adic numbers with zero integer part is a model of Dyson’s type. The traditional methods of quantum field theory such as Feynman diagrams, renormalization theory and Wilson’s renormalization group have analogs in the *p*-adic case. The main results of [1] are briefly outlined in Sect. 2.

On the other hand, there has been recently some interest on the possibility of a *p*-adic formulation of string theory (see [5–12]).

All this explains our interest in the Feynman amplitudes over the *p*-adic field. The remarkable feature of *p*-adic models is the exact representation of Feynman and string scattering amplitudes as a sum of elementary functions. Namely, let us consider a general Feynman amplitude over Q_p in coordinate representation,

$$F(x_v; v \in V_{\text{ext}}) = \int \prod_{v, v' \in V_{\text{ext}} \cup V_{\text{int}}} \|x_v - x_{v'}\|_p^{a(v, v')} \prod_{v \in V_{\text{int}}} dx_v, \quad (1.1)$$

where the integral taken over $Q_p^{|V_{\text{int}}|}$, p is a fixed prime number, Q_p is a *p*-adic field, $\|\cdot\|_p$ is a *p*-adic norm (further on, the sign p will be omitted), V_{ext} (V_{int}) is a set of external (internal) vertices, $a(v, v') \in \mathbb{C}$ for each pair $v \in V_{\text{ext}} \cup V_{\text{int}}$, $v' \in V_{\text{ext}} \cup V_{\text{int}}$ (we identify a pair (v, v') with (v', v)), dx is a Haar measure on Q_p , normalized such

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★★ This work was supported in part by the French Government

that

$$\int_{\|x\| \leq 1} dx = 1. \quad (1.2)$$

Every given vector of external variables $x = (x_v; v \in V_{\text{ext}}) \in Q_p^{|V_{\text{ext}}|}$ generates on V_{ext} a hierarchy A_x . We recall (see, for example [4, 12]) that a hierarchy A on a finite set V is a family of subsets of V , such that $V \in A$, $\{v\} \in A$ for every $v \in V$, and for each pair $V' \in A$, $V'' \in A$ either $V' \cap V'' = \emptyset$ or $V' \subset V''$, or $V'' \subset V'$. For every $V' \in A$ we denote by $\tau(V')$ the minimal set in A , which contains V' , but does not coincide with V' (we assume $V' \neq V$). In the following we shall consider only hierarchies such as

$$1 < |K(V')| \leq p, \quad V' \in A', \quad (1.3)$$

where

$$K(V') = \{V'' \in A \mid \tau(V'') = V'\}, \quad (1.4)$$

$$A' = \{V' \in A \mid |V'| > 1\}. \quad (1.5)$$

In our case, the hierarchy A_x on V_{ext} is defined as

$$A_x = \left\{ V \subset V_{\text{ext}} : \max_{\substack{v \in V \\ v' \in V}} \|x_v - x_{v'}\| < \min_{\substack{v \in V \\ v' \in V_{\text{ext}} \setminus V}} \|x_v - x_{v'}\| \right\}. \quad (1.6)$$

Note, that for every A on V_{ext} there exists $x \in Q_p^{|V_{\text{ext}}|}$ such that $A_x = A$.

Let

$$F_A(x) = \begin{cases} F(x), & \text{if } A_x = A \\ 0, & \text{otherwise} \end{cases}. \quad (1.7)$$

Then

$$F(x) = \sum_A F_A(x), \quad (1.8)$$

where the sum goes over all hierarchies on V_{ext} .

Let $A_x = A$. The main result of Sect. 3 is the following:

$$F_A(x) = \sum_{I(A)} \prod_{V \in A'} C(V, I) \max_{\substack{v \in V \\ v' \in V}} \|x_v - x_{v'}\|^{\lambda(V, I)}, \quad (1.9)$$

where the sum taken over all partitions of V_{int} , indexed by the elements of A' :

$$I(A) = \{I(V), V \in A'\}, \quad I(V') \cap I(V'') = \emptyset, \quad \text{if } V' \neq V'', \quad \left(\bigcup_{V \in A'} I(V) \right) = V_{\text{int}}, \quad (1.10)$$

$$\lambda(V, I) = a(V(I)) - \sum_{V' \in K(V)} a(V'(I)) + |I(V)|, \quad (1.11)$$

$$V(I) = \left(\bigcup_{V' \subseteq V} I(V') \right) \cup V, \quad (1.12)$$

$$C(V, I) = \int \prod_{v, v' \in I(V)} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in I(V)} \left(\prod_{V' \in K(V)} \|y_v - \alpha_{V'}\|^{a(v, V'(I))} \right) dy. \quad (1.13)$$

Integral (1.13) taken over $Q_p^{|I(V)|}$, $\{\alpha_{V'}, V' \in K(V)\}$ is an arbitrary set of p -adic numbers, such that $\|\alpha_{V'} - \alpha_{V''}\| = 1$, if $V' \neq V''$ (coefficient $C(V, I)$ does not depend on the choice of $\alpha = \{\alpha_{V'}, V' \in K(V)\}$). Here we use the notations

$$a(W, W') = \sum_{v \in W, v' \in W'} a(v, v'), \quad W, W' \subset V_{\text{ext}} \cup V_{\text{int}}, \quad (1.14)$$

$$a(W) = a(W, W). \quad (1.15)$$

We see that the integral representation for $C(V, I)$ is a generalization of the integral representation of string scattering amplitude in the Koba–Nielsen form.

Let

$$F_1 = \int \prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V} \prod_{i=0}^k \|y_v - \alpha_i\|^{a(v, i)} dy_v, \quad (1.16)$$

where $0 \leq k < p$, $\alpha = \{\alpha_i, i = 0, \dots, k\}$ is an arbitrary set of p -adic numbers, such that $\|\alpha_i - \alpha_j\| = 1$, if $i \neq j$, V is a finite set, $a(v, v') \in \mathbb{C}$ for each pair $v \in V, v' \in V$ (we identify (v, v') with (v', v)). In Sect. 4 we prove that the calculation of this integral can be reduced to that of the following one:

$$F_2 = \int \prod_{\substack{\|y_v\| < 1 \\ v \in V \setminus \{v_0\}}} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V \setminus \{v_0\}} dy_v, \quad (1.17)$$

where $y_{v_0} = 0$. We shall prove that

$$F_2(a_V) = p^{-|V|+1} \sum_A \prod_{V' \in A} \frac{1}{p^{a(V') + |V'| - 1} - 1} \cdot \frac{(p-1)!}{(p - |K(V')|)!}, \quad (1.18)$$

the sum taken over all hierarchies A on V .

2. P -Adic Scalar Models and Dyson's Hierarchical Models

Let a generalized random field $P(\phi)$ on the Q_p be given, i.e. a system of probability distributions $P\{(\phi, f_1), \dots, (\phi, f_m)\}$ with the usual conditions of accordance. Here $f_i = f_i(x)$, $i = 1, \dots, m$ are arbitrary test functions in the space $S(Q_p)$. $S(Q_p)$ is a space of locally piece-wise finite complex-valued functions on Q_p (see [15]).

We define the scaling operator

$$R_\lambda^a P(\phi) = P(\|\lambda\|^{1-(a/2)} \phi_\lambda), \quad (2.1)$$

where a is a real number, $1 \leq a \leq 2$, $\lambda \in Q_p$,

$$(\phi_\lambda, f) = (\phi(\lambda x), f(x)),$$

and $P(\|\lambda\|^{1-(a/2)} \phi_\lambda)$ is a generalized random field with probability distributions

$$P\{(\|\lambda\|^{1-(a/2)} \phi_\lambda, f_1), \dots, (\|\lambda\|^{1-(a/2)} \phi_\lambda, f_m)\}.$$

A generalized random field is scaling invariant if

$$R_\lambda^a P(\phi) = P(\phi) \quad (2.2)$$

for all $\lambda \in Q_p$. A generalized random field is called translation invariant if $P(\phi(x)) = P(\phi(x+a))$ for any $a \in Q_p$.

It is easy to see that a generalized gaussian random field with zero mean and binary correlation function

$$\langle \varphi(x_1), \varphi(x_2) \rangle = \text{const} \|x_1 - x_2\|^{a-2} \quad (2.3)$$

is a scaling and translation invariant random field.

By $\mathcal{D}_p$ we denote the set of all p -adic numbers with zero integer part. If

$$x = \sum_{i=n}^{\infty} a_i p^i, \quad 0 \leq a_i < p-1 \quad (2.4)$$

is a p -adic number, then $\{x\}$ denotes its fraction part

$$\{x\} = \sum_{i=n}^{-1} a_i p^i, \quad (2.5)$$

$$\mathcal{D}_p = \{x \in \mathcal{Q}_p : x = \{x\}\}. \quad (2.6)$$

$\mathcal{D}_p$ has a natural hierarchical structure, which consists from all sets of the type

$$V_i^n = \{x \in \mathcal{D}_p : \|p^n x - i\| \leq 1\}, \quad i \in \mathcal{D}_p, \quad n = 0, 1, 2, \dots \quad (2.7)$$

The discretization of a generalized random field φ on $\mathcal{D}_p$ is defined as a random field ξ on $\mathcal{D}_p$ such that

$$\xi = \{\xi_j = (\varphi, \chi_j), j \in \mathcal{D}_p\}, \quad (2.8)$$

where $\chi_j(x) = \chi(x-j)$, $\chi(x)$ is the characteristic function of the ball $Z_p = \{x \in \mathcal{Q}_p : \|x\| \leq 1\}$. The following operations on a random field $\xi = \{\xi_j, j \in \mathcal{D}_p\}$ are defined by the formulae

$$r_\lambda^a : \xi_j \rightarrow \xi'_j = \|\lambda\|^{-(a/2)} \sum_{i: \{i\lambda^{-1}\} = j, j \in \mathcal{D}_p} \xi_i, \quad \|\lambda\| \geq 1, \quad (2.9)$$

$$t_i : \xi_j \rightarrow \xi_{\{j+i\}}, \quad i \in \mathcal{D}_p. \quad (2.10)$$

If a generalized random field φ is a translation and scaling invariant, then its discretization is invariant relative to actions $t_i, i \in \mathcal{D}_p$ and $r_\lambda^a, \|\lambda\| > 1$. The gaussian scaling and translation invariant random field on $\mathcal{Q}_p$ also may be defined by the hamiltonian

$$H_0 = \frac{1}{2} \int \|x - y\|^{-a} \varphi(x) \varphi(y) dx dy. \quad (2.11)$$

One can show that the discretization of the random field with the hamiltonian (2.11) is a gaussian random field on $\mathcal{D}_p$ with the hamiltonian

$$\tilde{H}_0 = \frac{1}{2} \sum_{i,j \in \mathcal{D}_p} d(i,j) \xi_i \xi_j, \quad (2.12)$$

where

$$d(i,j) = \begin{cases} \|i-j\|^{-a}, & \text{if } i \neq j \\ \frac{p^{-a} - p^{1-a}}{1 - p^{1-a}}, & \text{otherwise} \end{cases}. \quad (2.13)$$

Note that the hamiltonian $\tilde{H}_0$ is the hamiltonian of the gaussian Dyson hierarchical model.

Further we shall have to deal with the hamiltonians in momentum representation. A hamiltonian in the ball $\Omega = \{k \in Q_p: \|k\| \leq R\}$ is an expression of the form

$$H(\sigma) = \sum_{m=1}^n \int_{\Omega^m} h_m(k_1, \dots, k_m) \delta(k_1 + \dots + k_m) \prod_{i=1}^m \sigma(k_i) dk_i. \quad (2.14)$$

A formal hamiltonian is a formal series in ε

$$H = H_0 + \varepsilon H_1 + \varepsilon^2 H_2 + \dots, \quad (2.15)$$

whose coefficients are finite-particle hamiltonians of the type (2.14). In what follows the coefficient H_0 will be fixed:

$$H_0 = \frac{1}{2} \int_{\Omega} \|k\|^{a-1} |\sigma(k)|^2 dk. \quad (2.16)$$

Wilson's renormalization transformation, as in a real case, is defined by the formula

$$R_{\lambda, \Omega}^a H = \ln \langle \exp(R_{\lambda}^a H(\sigma_0 + \sigma_1)) \rangle_{\mu(d\sigma_1)}, \quad (2.17)$$

where $(R_{\lambda}^a H)(\sigma) = H(\|\lambda\|^{-a/2} \sigma(\lambda^{-1}))$ is a hamiltonian of the random field in the ball $\lambda\Omega$, $\sigma(k)$ is a configuration in the ball $\lambda\Omega$, $\sigma(k) = \sigma_0(k) + \sigma_1(k)$, $\sigma_0(k) = \chi_{\Omega}(k)\sigma(k)$, $\sigma_1(k) = (\chi_{\lambda\Omega}(k) - \chi_{\Omega}(k))\sigma(k)$, $\chi_{\Omega}(k)$ is a characteristic function of the ball Ω , and the average taken with respect to the gaussian measure $\mu(d\sigma_1)$ with zero mean and binary correlation function

$$\langle \sigma_1(k)\sigma_1(k') \rangle = \delta(k + k')(\chi_{\lambda\Omega}(k) - \chi_{\Omega}(k))\|k\|^{1-a}. \quad (2.18)$$

The branch of gaussian fixed points of Wilson's renormalization group is determined by the hamiltonian $H_0 = H_0(a)$. An investigation of the spectrum of the differential of the renormalization group on this branch shows that $a_0 = \frac{3}{2}$ is the bifurcation point. One can try to construct a new branch of non-gaussian fixed points as power series in the deviation of the parameter a from the bifurcation value a_0 . As in the real case, (see [16]), we seek a solution in the class of projection hamiltonians, in the form

$$H = H(\sigma, \varepsilon) = \ln \langle \exp\{u(\varepsilon)\varphi^4(\sigma_0 + \sigma_1)\} \rangle_{\mu(d\sigma_1)}, \quad (2.19)$$

where

$$\varphi^4(\sigma) = \int \delta(k_1 + \dots + k_4) \prod_{i=1}^4 \sigma(k_i) dk_i, \quad (2.20)$$

σ_0 is a configuration in the ball Ω , and the average taken with respect to the gaussian measure with binary correlation function

$$\langle \sigma_1(k)\sigma_1(k') \rangle = \delta(k + k')(1 - \chi_{\Omega}(k))\|k\|^{1-a} = \delta(k + k')(1 - \chi_{\Omega}(k))\|k\|^{-(1/2)-\varepsilon}. \quad (2.21)$$

The only quantity which is not defined here is $u(\varepsilon)$, which we assume to be a formal power series in ε :

$$u(\varepsilon) = \sum_{j=1}^{\infty} u_j \varepsilon^j. \quad (2.22)$$

In the computation of a projection hamiltonian divergences appear. Namely, in

φ^4 theory with the propagator

$$(1 - \chi_\Omega(k)) \|k\|^{-(1/2) - \varepsilon} \quad (2.23)$$

the diagrams with two and four external lines have poles when $\varepsilon \rightarrow 0$. The theory of analytic renormalization analogous to the real case exists in the p -adic case. One can show that

$$\begin{aligned} R_{\Omega, \lambda}^a \text{A.R.} \ln \langle \exp \{u(\varepsilon) \varphi^4(\sigma_0 + \sigma_1)\} \rangle_{\mu(d\sigma_1)} \\ = \exp \left(\tau \beta(u) \frac{d}{du} \right) \text{A.R.} \ln \langle \exp \{u(\varepsilon) \varphi^4(\sigma_0 + \sigma_1)\} \rangle_{\mu(d\sigma_1)}, \end{aligned} \quad (2.24)$$

where A.R. denotes the analytic renormalization with minimal subtractions, $\tau = 2 \ln \|\lambda\|$,

$$\beta(u) = \varepsilon u + \sum_{n=2}^{\infty} c_n u^n, \quad (2.25)$$

and the coefficients c_n for $n \geq 2$ do not depend on ε .

The renormalization projection hamiltonian $\text{A.R.} \ln \langle \exp(u(\varepsilon) \varphi_4) \rangle_{\mu(d\sigma_1)}$ is invariant under the action of the renormalization group, if

$$\beta(u) = 0. \quad (2.26)$$

This equation has two solutions in the formal power in ε . The solution $u = 0$ corresponds to a gaussian fixed point. A nontrivial non-gaussian solution is obtained from the solution of

$$\varepsilon + \sum_{n=2}^{\infty} c_n u^{n-1} = 0. \quad (2.27)$$

It is easy to check that $c_2 \neq 0$ and therefore Eq. (2.27) is indeed solvable. This solution is a p -adic field-theoretical description of the non-trivial solution in Dyson's hierarchical model, which was investigated in [2–5].

3. Feynman Amplitudes in Coordinate Representation

First of all we shall prove that A_x , defined by (1.6) with the function

$$m_x(V) = \max_{\substack{v \in V \\ v' \in V}} \|x_v - x_{v'}\| \quad (3.1)$$

on it, is an indexed hierarchy. We recall that an indexed hierarchy on V is a pair (A, m) , where A denotes a given hierarchy on V and m is a positive function on A , satisfying the following conditions:

- 1) $m(V') = 0$ if and only if $|V'| = 1$,
- 2) if $V' \subset V''$, then $m(V') < m(V'')$.

In addition to these conditions we shall consider the functions $m(V)$ for which

$$m(V) = p^{n(V)}, \quad n(V) \in \mathbb{Z} \quad (3.2)$$

for every $V \in A$.

It suffices to show that if $V' \in A_x$, $V'' \in A_x$ and $V' \cap V'' \neq \emptyset$, then $V' \subset V''$ or $V'' \subset V'$. In fact, let $x \in V' \cap V''$, $y \in V' \setminus V''$, $z \in V'' \setminus V'$. From the definition of V' $\|x - y\| < \|y - z\|$ follows and hence, from the ultrametricity of the p -adic norm $\|y - z\| = \|x - z\|$, but this contradicts the definition of V'' .

It is easy to see that for every indexed hierarchy (A, m) on V_{ext} there exists $x = (x_v, v \in V_{\text{ext}})$ such that $(A_x, m_x) = (A, m)$. To every indexed hierarchy (A, m) corresponds a family of sets $\{\mathcal{D}(V, m), V \in A\}$, where

$$\mathcal{D}(V, m) = \{y \in Q_p : m(V) \leq \|y - x_v\| < m(\tau(V)), \forall v \in V\}. \quad (3.3)$$

We put $m(\tau(V_{\text{ext}})) = \infty$.

Let $\varphi: A \rightarrow V_{\text{ext}}$ be any function such that $\varphi(V) \in V$ for every $V \in A$.

Lemma 1. 1) If $V \in A$, $V' \in A$, $V \subset V'$, $V \neq V'$ and $y \in \mathcal{D}(V, m)$, $y' \in \mathcal{D}(V', m)$, then

$$\|y' - y\| = \|y' - x_{\varphi(V)}\|.$$

2) If $V \cap V' = \emptyset$, $V \in A$, $V' \in A$ and V'' is a minimal set in A , containing $V \cup V'$, $y \in \mathcal{D}(V, m)$, $y' \in \mathcal{D}(V', m)$, then

$$\|y' - y\| = m(V'').$$

3) $\{\mathcal{D}(V, m); V \in A\}$ is a partition of Q_p .

Proof. Let $y \in \mathcal{D}(V, m)$, $y' \in \mathcal{D}(V', m)$, $V \subset V'$, $V \neq V'$.

Then

$$\|y - x_{\varphi(V)}\| < m(\tau(V)) \leq m(V') \leq \|y' - x_{\varphi(V)}\|,$$

and therefore $\|y - y'\| = \|y' - x_{\varphi(V)}\|$. If $V \cap V' = \emptyset$, $y \in \mathcal{D}(V, m)$, $y' \in \mathcal{D}(V', m)$, then

$$\|y - x_{\varphi(V)}\| < m(\tau(V)) \leq m(V'') = \|x_{\varphi(V)} - x_{\varphi(V')}\|,$$

$$\|y' - x_{\varphi(V')} \| < m(\tau(V')) \leq m(V''),$$

and hence

$$\|y' - x_{\varphi(V)}\| = \|y - y'\| = m(V'').$$

Let $V \neq V'$ and $y \in \mathcal{D}(V, m) \cap \mathcal{D}(V', m)$. If $V \subset V'$, $V \neq V'$, then

$$m(V) \leq \|y - x_{\varphi(V)}\| < m(\tau(V)) \leq \|y - x_{\varphi(V')}\|,$$

but this contradicts the first part of the lemma. If $V \cap V' = \emptyset$, then

$$\|x_{\varphi(V)} - x_{\varphi(V')}\| < \max(\|y - x_{\varphi(V)}\|, \|y - x_{\varphi(V')}\|) < m(V'').$$

At last

$$\bigcup_{V \in A} \mathcal{D}(V, m) = Q_p.$$

Lemma 1 is proved.

Everywhere below we have used the next notations: $a_V = (a(v, v'))_{v \in V}^{v' \in V}$ is a matrix, $b_V = (b(v))_{v \in V}$ is a vector,

$$a(V, V') = \sum_{v \in V, v' \in V'} a(v, v'), \quad (3.4)$$

$a(V) = a(V, V)$, $a_V^{V'} = (a(v, V'))_{v \in V}$ is a vector, $a(v, v') \in \mathbb{C}$, $b'(v) \in \mathbb{C}$ for every $v \in V$, $v' \in V$.

Let r_1, r_2 be real numbers, $\infty \geq r_2 > r_1 \geq 0$. Denote by

$$f(r_1, r_2; a_V, b_V) = \int_{r_1 \leq \|y_v\| < r_2, v \in V} \prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V} \|y_v\|^{b(v)} dy_v. \quad (3.5)$$

Lemma 2.

$$f(r_1, r_2; a_V, b_V) = \sum_{V_1 \cup V_2 = V} f(r_1, \infty; a_{V_1}, b_{V_1}) f(0, r_2; a_{V_2}, b_{V_2} + a_{V_2}^{V_1}) \quad (3.6)$$

where the sum goes over all partitions $\{V_1, V_2\}$ of V .

Proof. We prove this lemma by induction with respect to the number of elements of V . For $|V| = 1$ this formula is checked directly:

$$\begin{aligned} \int_{r_1 \leq \|y\| < r_2} \|y\|^b dy &= \int_{r_1 \leq \|y\| < r_2} \|y\|^b dy + \int_{\|y\| < r_1} \|y\|^b dy + \int_{r_1 \leq \|y\|} \|y\|^b dy \\ &= \int_{\|y\| < r_2} \|y\|^b dy + \int_{r_1 \leq \|y\|} \|y\|^b dy \end{aligned}$$

(we used $\int_{\mathbb{Q}_p} \|y\|^b dy = 0$).

Note that

$$f(r_1, r_2; a_V, b_V) = f(0, r_2; a_V, b_V) - \sum_{V_1 \subseteq V, V_1 \neq \emptyset} f(0, r_1; a_{V_1}, b_{V_1}) f(r_1, r_2; a_{V_2}, b_{V_2} + a_{V_2}^{V_1}). \quad (3.7)$$

Applying the assumption of induction to $f(r_1, r_2; a_{V_2}, b_{V_2} + a_{V_2}^{V_1})$, we have

$$\begin{aligned} f(r_1, r_2; a_V, b_V) &= f(0, r_2; a_V, b_V) - \sum_{V_1 \subseteq V, V_1 \neq \emptyset} (f(0, r_1; a_{V_1}, b_{V_1}) \\ &\quad \cdot \sum_{V_2 \subseteq V \setminus V_1} f(r_1, \infty; a_{V_2}, b_{V_2} + a_{V_2}^{V_1}) f(0, r_2; a_{V_3}, b_{V_3} + a_{V_3}^{V_1} + a_{V_3}^{V_2})), \end{aligned} \quad (3.8)$$

where $V_3 = V \setminus (V_1 \cup V_2)$. By denoting $V_4 = V_1 \cup V_2$ and changing the order of the summation, we get

$$\begin{aligned} f(r_1, r_2; a_V, b_V) &= f(0, r_2; a_V, b_V) - \sum_{V_4 \subseteq V, V_4 \neq \emptyset} \left\{ \left(\sum_{V_1 \subseteq V_4, V_1 \neq \emptyset} f(0, r_1; a_{V_1}, b_{V_1}) \right. \right. \\ &\quad \left. \left. : f(r_1, \infty; a_{V_2}, b_{V_2} + a_{V_2}^{V_1}) f(0, r_2; a_{V_3}, b_{V_3} + a_{V_3}^{V_4}) \right) \right\}. \end{aligned} \quad (3.9)$$

But

$$\sum_{V_1 \subseteq V_4} f(0, r_1; a_{V_1}, b_{V_1}) f(r_1, \infty; a_{V_2}, b_{V_2} + a_{V_2}^{V_1}) = f(0, \infty; a_{V_4}, b_{V_4}) = 0, \quad (3.10)$$

and hence

$$\sum_{V_1 \subseteq V_4, V_1 \neq \emptyset} f(0, r_1; a_{V_1}, b_{V_1}) f(r_1, \infty; a_{V_2}, b_{V_2} + a_{V_2}^{V_1}) = -f(r_1, \infty; a_{V_4}, b_{V_4}), \quad (3.11)$$

where the summation is over $V_1 \neq \emptyset$, $V_1 \subset V_4$. Therefore,

$$f(r_1, r_2; a_V, b_V) = f(r_1, r_2; a_V, b_V) + \sum_{V_2 \subseteq V, V_4 \neq \emptyset} f(r_1, \infty; a_{V_4}, b_{V_4}) f(0, r_2; a_{V_3}, b_{V_3} + a_{V_3}^{V_4}), \quad (3.12)$$

$V_3 = V \setminus V_4$. Lemma 2 is proved.

Theorem 1. *Let $A_x = A$. Then*

$$F(x) = \sum_{I(A)} \prod_{V \in A'} C(V, I) \max_{\substack{v \in V \\ v' \in V}} \|x_v - x_{v'}\|^{s(V, I)}, \quad (3.13)$$

where

$$C(V, I) = \int \prod_{v', v \in I(V)} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in I(V)} \left(\prod_{V' \in K(V)} \|y_v - \alpha_{V'}\|^{a(v, V'(I))} dy_v \right), \quad (3.14)$$

$\{\alpha_{V'}, V' \in K(V)\}$ is an arbitrary set of p-adic numbers, such that $\|\alpha_{V'} - \alpha_{V''}\| = 1$ if $V' \neq V''$ (for example, $\alpha = \{\alpha_{V'}; V' \in K(V)\} = \{0, 1, \dots, |K(V)| - 1\}$). $C(V, I)$ do not depend on the choice of α . For the other notations see Sect. 1.

Proof. Let $A_x = A$ and $\varphi: A \rightarrow V_{\text{ext}}$ is any function such that $\varphi(V) \in V, V \in A$. Let $T(A) = \{T(V), V \in A\}$ is a partition of V_{int} , indexed by elements of A . Denote by

$$g(V, T, x) = \int_{\substack{y_v \in \mathcal{S}(V, m_x) \\ v \in T(V)}} \prod_{v, v' \in T(V)} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in T(V)} \|y_v - x_{\varphi(V)}\|^{a(v, V(T))} dy_v, \quad (3.15)$$

where $V(T) = \left(\bigcup_{V' \subset V} T(V') \right) \cup V$.

From Lemma 1 it is easy to derive that

$$F(x) = \sum_{T(A)} \prod_{V \in A'} g(V, T, x) m_x(V)^{s(V, T)}, \quad (3.16)$$

where the sum goes over all partitions of V_{int} , indexed by elements of A ,

$$s(V, T) = \sum_{\substack{V' V'' \in K(V) \\ V' \neq V''}} a(V'(T) \cup T(V'), V''(T) \cup T(V'')). \quad (3.17)$$

According to Lemma 2

$$g(V, T, x) = \sum_{T_1(V) \cup T_2(V) = T(V)} f(m(V), \infty; a_{T_1(V)}, a_{T_1(V)}^{V(T)}) f(0, m(\tau(V); a_{T_2(V)}, a_{T_2(V)}^{V(T) + T_1(V)}), \quad (3.18)$$

the sum goes over all partitions $(T_1(V), T_2(V))$ of $T(V)$.

As $f(0, \infty; a_V, b_V) = 0$ for any a_V, b_V , we may write

$$F(x) = \sum_{I(A)} \prod_{V \in A'} \left(\sum_{\substack{(\bigcup_{V' \in K(V)} T_2(V')) \cup T_1(V) = I(V)}} f(m(V), \infty; a_{T_1(V)}, a_{T_1(V)}^{V(I)T_1(V)}) \cdot \prod_{V' \in K(V)} f(0, m(V); a_{T_2(V)}, a_{T_2(V)}^{V'(I)}) \right) m_x(V)^{s(V, I)}, \quad (3.19)$$

where the sum goes over all partitions of V_{int} , indexed by elements of A' , the internal sum taken over all partitions of

$$I(V) = \left(\bigcup_{V' \in K(V)} T_2(V') \right) \cup T_1(V), \quad V(I) = \left(\bigcup_{V' \subseteq V} I(V') \right) \cup V, \\ s(V, I) = \sum_{\substack{V', V'' \in K(V) \\ V' \neq V''}} a(V'(I), V''(V)). \quad (3.20)$$

As

$$\left(\bigcup_{V' \in K(V)} \{y \in Q_p: \|y - x_{\varphi(V')}\| < m(V)\} \right) \cup \{y \in Q_p: \|y - x_{\varphi(V)}\| \geq m(V)\} = Q_p, \quad (3.21)$$

we get

$$F(x) = \sum_{I(A)} \prod_{V \in A'} m_x(V)^{s(V,I)} \cdot \left(\int \prod_{v, v' \in I(V)} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in I(V)} \left(\prod_{V' \in K(V)} \|y_v - x_{\varphi(V')}\|^{a(v, V'(I))} \right) dy_v \right). \quad (3.22)$$

Performing a change of variables,

$$y_v = y_v p^{-n}, \quad v \in I(V), \quad (3.23)$$

$$\alpha_{V'} = x_{\varphi(V')} p^{-n}, \quad V' \in K(V), \quad (3.24)$$

where $\|p^n\| = m_x(V)$, we obtain

$$F(x) = \sum_{I(A)} \prod_{V \in A'} C(V, I, \alpha) m_x(V)^{\lambda(V, I)}, \quad (3.25)$$

$$C(V, I, \alpha) = \int \prod_{v, v' \in I(V)} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in I(V)} \left(\prod_{V' \in K(V)} \|y_v - \alpha_{V'}\|^{a(v, V'(I))} \right) dy, \quad (3.26)$$

$$\lambda(V, I) = a(V(I)) - \sum_{V' \in K(V)} a(V'(I)) + |I(V)|. \quad (3.27)$$

Note that $\|\alpha_{V'} - \alpha_{V''}\| = 1$ for each pair $V', V'' \in K(V)$ $V' \neq V''$. In the next section we shall see that $C(V, I, \alpha)$ does not depend on the choice of α .

Note also that $C(V, I)$ is a Feynman amplitude of the contracted graph

$$G_V|_{\{G_{V'}, V' \in K(V)\}}.$$

Here G_V is a graph with the set of vertices $V(I)$. To each external vertex $\{V'\}$ of this contracted graph we must assign an external variable $\alpha_{V'}$.

$\lambda(V, I)$ also has a simple geometrical description.

4. The Calculation of Coefficients $C(V, I)$ and String Amplitudes in the Koba–Nielsen Form

Let us consider an integral

$$F_1 = \int \prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V} \left(\prod_{i=0}^k \|y_v - \alpha_i\|^{b_i(v)} \right) dy_v, \quad (4.1)$$

where $1 \leq k \leq p-1$, $\|\alpha_i - \alpha_j\| = 1$ if $i \neq j$. This integral is a generalization of Koba–Nielsen amplitude (some examples of this amplitude were calculated in [10, 11]).

We introduce the next partition of Q_p :

$$\mathcal{Q}_i = \{y \in Q_p: \|y - \alpha_i\| < 1\}, \quad i = 0, 1, \dots, k, \quad (4.2)$$

$$\mathcal{Q}_j = \{y \in Q_p: \|y - \alpha_0\| = 1, (y - \alpha_0) \bmod p = j\}, \quad j \in J. \quad (4.3)$$

where $J = \{0, 1, \dots, p-1\} \setminus \{(\alpha_i - \alpha_0) \bmod p, i=0, \dots, k\}$. As $|J| = p - k - 1$, it is convenient to renumerate the family $\mathcal{D}_j, j \in J$ by $i = k+1, \dots, p-1$. Finally,

$$\mathcal{D}_p = \{y \in \mathcal{Q}_p : \|y - \alpha_0\| > 1\}. \quad (4.4)$$

As above, it is easy to show that $\{\mathcal{D}_i, i=0, \dots, p\}$ is a partition of $\mathcal{Q}_p$. Therefore

$$\begin{aligned} F_1 &= \sum_{\bigcup_{i=0}^p V_i = V} \prod_{i=0}^k \int_{\substack{v, v' \in \mathcal{D}_i \\ v \in V_i}} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_i} \|y_v - \alpha_i\|^{b_i(v)} dy_v \\ &\cdot \prod_{j=k+1}^{p-1} \int_{\substack{y_v \in \mathcal{D}_j \\ v \in V_j}} \prod_{v, v' \in V_j} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_j} dy_v \\ &\cdot \int_{\substack{y_v \in \mathcal{D}_p \\ v \in V_p}} \prod_{v, v' \in V_i} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_i} \|y_v - \alpha_0\|^{\sum_{i=0}^k b_i(v) + a(v, V \setminus V_p)} dy_v. \end{aligned} \quad (4.5)$$

But for $i=0, \dots, k$,

$$\begin{aligned} &\int_{\substack{y_v \in \mathcal{D}_i \\ v \in V_i}} \prod_{v, v' \in V_i} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_i} \|y_v - \alpha_i\|^{b_i(v)} dy_v \\ &= \int_{\substack{\|y_v\| < 1 \\ v \in V_i}} \prod_{v, v' \in V_i} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_i} \|y_v - \alpha_i\|^{b_i(v)} dy_v, \end{aligned} \quad (4.6)$$

for $i = k+1, \dots, p-1$

$$\int_{\substack{y_v \in \mathcal{D}_i \\ v \in V_i}} \prod_{v, v' \in V_i} \|y_v - y_{v'}\|^{a(v, v')} dy_v = \int_{\substack{\|y_v\| < 1 \\ v \in V_i}} \prod_{v, v' \in V_i} \|y_v - y_{v'}\|^{a(v, v')} dy_v, \quad (4.7)$$

and

$$\begin{aligned} &\int_{\substack{y_v \in \mathcal{D}_p \\ v \in V_p}} \prod_{v, v' \in V_p} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_p} \|y_v - \alpha_0\|^{\sum_{i=0}^k b_i(v) + a(v, V \setminus V_p)} dy_v \\ &= \int_{\substack{y_v < 1 \\ v \in V_p}} \sum_{v, v' \in V_p} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_p} \|y_v\|^{\sum_{i=0}^k b_i(v) + a(v, V) + 2} dy_v. \end{aligned} \quad (4.8)$$

In the last reduction we used the change of variables $y_v \rightarrow (1/y_v - \alpha_0)$, $v \in V_p$. Therefore, we may rewrite

$$F_1 = \sum_{\bigcup_{i=0}^p V_i = V} \prod_{i=0}^p g(a_{V_i}, b_i), \quad (4.9)$$

where

$$g(a_{V_i}, b_i) = \int_{\substack{\|y_v\| < 1 \\ v \in V_i}} \prod_{v, v' \in V_i} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V_i} \|y_v\|^{b_i(v)} dy_v, \quad (4.10)$$

$$b_i(v) = 0, \quad v \in V_i, \quad i = k+1, \dots, p-1, \quad (4.11)$$

$$b_p(v) = -\left(\sum_{i=0}^k b_i(v) + a(v, V) + 2\right), \quad v \in V_p. \quad (4.12)$$

So, we must calculate the integral of the type

$$F_2(a_V) = \int_{\substack{|y_v| < 1 \\ v \in V \setminus \{v_0\}}} \prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \prod_{v \in V \setminus \{v_0\}} dy_v, \quad (4.13)$$

where v_0 is a fixed element of V , $y_{v_0} = 0$, $a(v, v') \in \mathbb{C}$ for every pair (v, v') (we identify each pair (v, v') with (v', v)), $a(v, v) = 0$ for every $v \in V$ (in our case $V \setminus \{v_0\} = V_i$, $a(v, v_0) = b_i(v)$, $i = 0, 1, \dots, p$).

Lemma 3. *Let (A, m) be an indexed hierarchy on V , and*

$$\chi_{A, m}(y_v; v \in V \setminus \{v_0\}) = \begin{cases} 1, & \text{if } (A_y, m_y) = (A, m) \\ 0, & \text{otherwise} \end{cases}, \quad (4.14)$$

where $y = \{y_v, v \in V\}$, $y_{v_0} = 0$. Then

$$\int \chi_{A, m}(y_v; v \in V \setminus \{v_0\}) \prod_{v \in V \setminus \{v_0\}} dy_v = \prod_{V' \in A'} \left(\frac{m(V')}{p} \right)^{|K(V')| - 1} \frac{(p-1)!}{(p - |K(V)|)!}. \quad (4.15)$$

Proof of Lemma 3. Let T be a tree with the set of vertices V and the set of lines $L = \{l = (\varphi(V'), \varphi(V'')) | V' \in A, V'' \in K(V')\}$. Here $\varphi: A \rightarrow V$ is any function such that $\varphi(V') \in V'$ for every $V' \in A$, $\varphi(V) = v_0$. To each line $l = (\varphi(V'), \varphi(V''))$ we assign a variable $S_l = y_{\varphi(V')} - y_{\varphi(V'')}$. Then for every $y_v = S_{l_1} + S_{l_2} + \dots + S_{l_j}$, where $\{l_1, \dots, l_j\}$ is a path joining v with v_0 . It is clear that jacobian of this change of variables is equal to 1. And moreover, $(A_y, m_y) = (A, m)$ if and only if

- 1) $\|S_l\| = m(V') = p^{n(V')}$ for every $l = (\varphi(V'), \varphi(V''))$, $V' \in A'$, $V'' \in K(V')$.
- 2) $S_{l_1} p^{n(V')} \bmod p \neq S_{l_2} p^{n(V')} \bmod p$ for each pair $l_1 = (\varphi(V'), \varphi(V_1))$,

$l_2 = (\varphi(V'), \varphi(V_2))$, $V' \in A'$, $V_1 \in K(V')$, $V_2 \in K(V')$, $V_1 \neq V_2$.

Using

$$\int_{\substack{S_l = m(V') \\ sp^{n(V')} \bmod p = r}} ds = \frac{m(V')}{p}, \quad (4.16)$$

$r = 1, 2, \dots, p-1$, we obtain the formula (4.15).

Theorem 2. *Let $a(V') + |V'| - 1 > 0$ for every $V' \subseteq V$, $|V'| > 1$. Then*

$$F_2(a_V) = p^{-|V|+1} \sum_A \prod_{V' \in A'} \frac{1}{p^{a(V') + |V'| - 1} - 1} \cdot \frac{(p-1)!}{(p - |K(V)|)!}, \quad (4.17)$$

the sum goes over all hierarchies A on V .

Proof.

$$g(a_V) = \sum_{(A, m): m(V) < 1} \int \prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \chi_{A, m}(y_v; v \in V \setminus \{v_0\}) \prod_{v \in V \setminus \{v_0\}} dy_v, \quad (4.18)$$

where the sum taken over all indexed hierarchies on V , such that $m(V) < 1$. Note that this sum is equal to the double sum

$$\sum_A \sum_{m: m(V) < 0} \int \prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \chi_{A, m}(y_v; v \in V \setminus \{v_0\}) \prod_{v \in V \setminus \{v_0\}} dy_v, \quad (4.19)$$

where the external sum goes over all hierarchies on V and the internal sum goes over all set of integer numbers $\{n(V'), V' \in A'\}$, $n(V') \in \mathbb{Z}$, $m(V') = p^{n(V')}$, $n(V') < n(\tau(V'))$ for every $V' \in A'$ and $n(V) < 0$.

Let

$$a'(V') = a(V') - \sum_{V'' \in K(V')} a(V''), \quad V' \in A'. \quad (4.20)$$

Obviously, that

$$\prod_{v, v' \in V} \|y_v - y_{v'}\|^{a(v, v')} \chi_{A, m}(y_v; v \in V \setminus \{v_0\}) = \prod_{V' \in A'} m(V')^{a'(V')} \chi_{A, m}(y_v; v \in V \setminus \{v_0\}).$$

Note that if $\beta > 0$, then

$$\sum_{n < m} p^{n\beta} = \frac{p^{m\beta}}{p^{\beta-1}}. \quad (4.21)$$

Using (4.21) and Lemma 3, we obtain

$$\begin{aligned} \sum_A \sum_{n: n(V) < 0} \sum_{V' \in A'} p^{n(V')(a'(V') + |K(V')| + 1)} p^{-|K(V')| + 1} \cdot \frac{(p-1)!}{(p - |K(V')|)!} \\ = \sum_A \prod_{V' \in A'} p^{-|K(V')| + 1} \cdot \frac{(p-1)!}{(p - |K(V')|)!} \frac{1}{p^{\beta(V') - 1}}. \end{aligned} \quad (4.22)$$

Here we used that

$$\beta(V') = \sum_{V'' \subseteq V', V'' \in A'} (a'(V'') + |K(V'')| - 1) > 0. \quad (4.23)$$

As

$$\sum_{V'' \subseteq V', V'' \in A'} a'(V'') = a(V'), \quad (4.24)$$

$$\sum_{V'' \subseteq V', V'' \in A'} (|K(V'')| - 1) = |V'| - 1, \quad (4.25)$$

we have

$$\beta(V') = a(V') + |V'| - 1 > 0, \quad V' \subseteq V \quad (4.26)$$

by the assumption of the theorem. Finally we get

$$F_2(a_V) = p^{-|V|+1} \sum_A \prod_{V' \in A'} \frac{1}{p^{a(V') + |V'| - 1}} \cdot \frac{(p-1)!}{(p - |K(V')|)!}. \quad (4.27)$$

The theorem is proved.

This last formula defines also an analytic continuation of $F_2(a_V)$ from the domain (4.29) to the whole complex plane as a meromorphic function of $a(v, v')$, $v, v' \in V$.

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Communicated by K. Gawedzki

Received June 21, 1988; in revised form August 3, 1988